

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

School of Computing — Department of Computer Science & Engineering

Student Name: Sautrik Roy	Reg No: RA2511003010052
Course: 21CSC203P — Advanced Programming Practice	Department: Computer Science and Engineering

Unit I | Session 3 | SLO 1: Structured Programming & Bohm-Jacopini (Code: 1031)

Q: 1. Conceptual: Explain the three control structures identified in the Böhm-Jacopini theorem with simple Java examples.

Ans: The Böhm-Jacopini theorem proved that any computable function can be implemented using only three primary control structures:

1. **Sequence:** Ordered, step-by-step linear execution.
2. **Selection:** Conditional branching between paths (e.g., if-else or switch).
3. **Iteration:** Repetitive loop execution while a boolean predicate holds true (e.g., while, for).

```
// 1. Sequence
int a = 10;
int b = 20;
int sum = a + b;

// 2. Selection
if (sum > 25) {
    System.out.println("Sum is large");
} else {
    System.out.println("Sum is normal");
}

// 3. Iteration
for (int i = 0; i < 3; i++) {
    System.out.println("Iteration: " + i);
}
```

Q: 2. Analytical: Why did the Böhm-Jacopini theorem lead to the decline of the goto statement in modern programming languages?

Ans: Explanation: Prior to structured programming, code heavily relied on arbitrary jumps (goto), resulting in unstructured 'spaghetti code' that was impossible to trace, debug, or mathematically verify. Böhm-Jacopini provided the mathematical proof that goto is completely unnecessary, directly inspiring structured control statements with single-entry and single-exit points.

Q: 3. Application: Rewrite the following pseudocode (which uses goto) using structured constructs:

start:

x = x + 1

if (x < 10) goto start

Ans: Structured Equivalent in Java:

This construct executes the body at least once, making a do-while loop the most idiomatic equivalent:

```
int x = 0;
do {
    x = x + 1;
} while (x < 10);

// Alternatively using a standard while loop:
while (x < 10) {
    x = x + 1;
}
```